MIL-L-22569A(SH)
16 June 1977
SUPERSEDING
MIL-L-22569(SHIPS)
29 December 1960
(See 6.5)

MILITARY SPECIFICATION

# LIGHT, ULTRA-VIOLET, FLUORESCENT, PORTABLE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers a lightweight, portable, electrically-operated, ultraviolet, fluorescent light in a carrying case.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
J-C-175 - Cable, Assembly, Power, Electrical; For 125-Volt Equipment.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS LABORATORIES, INC.
Recognized Component Index

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 1285 Walt Whitman Road, Melville, L. I., New York 11746.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2.1), the contractor shall furnish a sample unit for first article inspection and approval (see 4.3 and 6.3).

3.2 Material. Material shall be as specified herein and shall meet the performance requirements of this specification.

3.3 Design and construction. The light shall consist of a housing containing a lamp, ballast, switch, wiring and connections necessary to operate the fluorescent type lamp specified in 3.3.6. A carrying case shall be provided for protection and storage. The length of the housing, including switch, shall not exceed 10 inches, the width shall not exceed 3 inches, and the height, including handle, shall not exceed 4 inches.

3.3.1 Housing. The housing shall be constructed of 22 gage cold rolled steel and shall accommodate a push switch, ballast, wiring and lamp.

3.3.2 Handle. The housing shall be equipped with a steel carrying handle, spot welded to the lamp housing on the top opposite the lamp.

3.3.3 Reflector. The housing shall have a built-in lamp reflector at least 8-7/8 inches in length, constructed of highly polished, specular finish, 0.030 inch thick aluminum. The reflector shall be so shaped as to concentrate ultra-violet energy into a 67-degree beam.

3.3.4 Switch. The switch shall be a sequence starting type, rated at 3 amperes, 125 volts (V), and shall be mounted at one end of the lamp housing.

3.3.4.1 Cycling. The switch shall be capable of withstanding 5,000 cycles of on-off (or start-stop) action at rated current and voltage without failure.

3.3.4.2 Dielectric. The switch shall be capable of withstanding a dielectric test voltage of 1250 V, 60 hertz (Hz) between current carrying parts and between current carrying parts and ground for 1 minute.

3.3.5 Cable assembly. The light shall be equipped with a three-wire cord, type SJO-18/3 at least 30 feet in length with a molded two-pole, three-wire, grounding type plug. The cable assembly shall be style 1 in accordance with J-C-175, and shall be secured by a strain relief bushing to prevent chafing.

3.3.5.1 Insulation resistance. The insulation resistance of the cable assembly shall be not less than 50,000 ohms when measured with a 500-volt direct current (V d.c.), indicating instrument between current carrying and non-current carrying parts.

3.3.5.2 Dielectric. The light interiors shall be capable of withstanding for 1 minute, a test voltage of 1250 V applied between points of opposite polarity and between live parts and ground at a frequency of 60 Hz.

3.3.6 Lamp. The lamp shall be a 6-watt (W) bi-pin base, 9-inch fluorescent lamp. The lamp shall be black light, fluorescent type, industry number F6T5/BL.

3.3.7 Filter. The lamp shall be surrounded with a tubular filter, together with necessary molded neoprene adapters, to provide maximum transmission of ultra-violet light, with maximum absorption of visible light. The filter shall be easily removable by hand, without damage to lamp or filter.

3.3.8 Carrying case. The carrying case, including cover shall be constructed of 22-gage steel and have rounded corners. The case shall house and incorporate the power cable, light and required repair parts.

3.3.8.1 Carrying case cover. The cover shall be attached to the carrying case by hinges and shall be provided with a smooth acting luggage catch and a carrying handle. The handle shall be a movable type to facilitate storage of the carrying case.



Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

3.3.9 Finish. The lamp housing and carrying case external and internal exposed surfaces shall be finished with a baked black crackle enamel on a compatible coat of primer.

3.3.10 Repair parts. The repair parts required for the light shall be as follows:

(a) Lamp; Industry number F6T5/BL (two required).
(b) Filter, black light, tubular shape, together with necessary molded neoprene adapters (one required).

3.3.11 Ruggedness. The carrying case, complete with light, cable assembly and repair parts shall be capable of withstanding the ruggedness test specified in 4.6.4 without any damage or distortion which might impair its serviceability.

3.4 Safety requirements. All wiring, mounting of connectors and bushings, strain relief of cords, splicing, spacing, exposure of live parts, insulation from shock hazard, and temperature rise during operation shall meet fire and safety requirements as approved by Underwriters' Laboratories, Inc.

3.5 Effectiveness of enclosure. The light shall be dripproof. There shall be no entry of water when tested as specified in 4.6.5. Gaskets may be employed to achieve enclosure effectiveness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 and 4.6.

4.3.1 First article inspection report. The contractor shall prepare a first article inspection report in accordance with the data ordering document included in the contract (see 6.2.2).

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all lights manufactured from the same batch and presented for inspection at the same time.

4.4.2 Sampling for examination and tests. A random sample of lights shall be selected from each lot in accordance with MIL-STD-105, Inspection Level II and subjected to the examination of 4.5, the insulation resistance test of 4.6.1, and the dielectric test of 4.6.2. Any light containing one or more defects and any light failing in one or more tests shall not be offered for delivery. An Acceptable Quality Level (AQL) of 2.5 shall apply.

4.5 Visual and dimensional examination. Each of the sample lights selected in accordance with 4.4.2 shall be examined to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Insulation resistance. Insulation resistance shall be measured between all current carrying parts and also between current carrying and noncurrent carrying parts, with a 500 V d.c. insulation resistance indicating instrument. The insulation resistance shall not be less than 50,000 ohms. During first article inspection, the test shall be conducted before and after the ruggedness test specified in 4.6.4.



Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

4.6.2 Dielectric test. Lighting fixture interiors shall be subjected for 1 minute to a dielectric test voltage, the effective potential of which is twice rated voltage plus 1,000 V applied between points of opposite polarity and also between live parts and ground. The frequency of the test voltage shall be 60 Hz. During first article inspection, the test shall be conducted before and after the ruggedness test specified in 4.6.4.

4.6.3 Switch test.

4.6.3.1 The switch shall be cycled 5000 cycles of start action and off action at rated voltage and current without failure.

4.6.3.2 The switch shall be subjected to a dielectric test voltage of 1250 V, 60 Hz, between current carrying parts and between current carrying parts and ground for 1 minute. The source of the test potential shall have a rating of at least 1 kilowatt (kW).

4.6.4 Ruggedness. The carrying case, complete with light, cable assembly, and repair parts, shall be dropped once from a height of 30 inches onto a rigidly mounted steel plate or concrete surface. The case, light housing and repair parts shall be examined for damage or distortion to determine compliance with 3.3.11.

4.6.5 Effectiveness of enclosure. The light, assembled in its operational configuration shall be subjected to the dripproof (15 degrees) test of MIL-STD-108. The light shall be examined for evidence of moisture.

4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Complete light. Each carrying case, complete as specified in 3.3.8, shall be unit protected and packaged in a snug fitting fiberboard container conforming to PPP-B-636, class weather-resistant with selection of the grade, variety and style at the option of the contractor. Container closure shall conform to method V as specified in the appendix to the container specification.

5.1.2 Level C. Each carrying case, complete as specified in 3.3.8, shall be preserved-packaged to afford protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when such meets the requirements of their level.

5.2 Packing. Lights packaged as specified in 5.1 shall be packed level A, B or C as specified (see 6.2.1) in accordance with MIL-E-17555.

5.3 Repair parts shall be preserved-packaged, packed and marked in accordance with MIL-F-17555 as applicable for the intended end use and destination for the level specified as follows:

| | Levels | |
|---|---|---|
| Destination | Preservation-packaging | Packing |
| Onboard | A | C |
| Stock | A | B |
| Immediate use | C | C |

5.4 Marking. In addition to any special marking required by the contract (see 6.2.1), interior packages and exterior shipping containers shall be marked for shipment in accordance with the requirements of MIL-E-17555.



Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The light covered by this specification is intended for use during tests to locate leaks in condensers.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required for inspection and approval (see 3.1, 4.3, and 6.3).
(c) Level of preservation-packaging and packing required (see 5.1, 5.2, and 5.3).
(d) Special marking, if required (see 5.4).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

| Paragraph | Data Requirement | Applicable DID |
|---|---|---|
| 4.3.1 | First article inspection report | UDI-T-23790 |

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article inspection.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6240-N285)



Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the currentversion before use.

FOLD

COMMANDER
NAVAL SHIP ENGINEERING CENTER (SEC 6124)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID

DEPARTMENT OF THE NAVY

DOD 316

COMMANDER
NAVAL SHIP ENGINEERING CENTER (SEC 6124)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20362

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-19T12:58Z
Check the source to verify that this is the current version before use.